Leaf Stomata Microscope Detective

Count the breathing pores and rank leaves by how they manage water.

Win condition: Most accurate detective ranking with clean data and strong evidence.

THE SCIENCE YOU ARE USING

Stomata: pores for gas exchange. Leaves breathe through tiny pores called stomata. They open to take in carbon dioxide and let oxygen out, but open stomata also lose water. The number and spacing of stomata reflect how a plant balances feeding itself against drying out.

Sampling and counting. You make a safe leaf peel or use prepared slides, count stomata in several fields of view, and average. Comparing counts across leaves lets you rank them by water strategy with real data.

HOW TO BUILD AND RUN IT

01 **Make a safe peel.** Paint clear polish on the leaf underside, let it dry, and lift the print with clear tape. Or use a prepared slide.

02 **Focus the field.** Place the peel under the microscope and focus until the stomata are sharp.

03 **Count several fields.** Count stomata in three or more fields of view and average to reduce error.

04 **Compare leaves.** Repeat for different leaves and build a team data table of counts per field.

05 **Rank by water strategy.** Use your counts to rank leaves from water-saving to water-spending, citing the data.

Safety: Clear nail polish gives off fumes, so it is applied only at the ventilated instructor station, kept away from heat, and capped when not in use. Gloves are available there if you do not want polish on your fingers; no goggles are needed. Handle glass slides by the edges. Plant or latex allergy: use the prepared slides instead and latex-free gloves.

MATERIALS

Leaf samples	varied
Clear tape	shared
Clear nail polish (ventilated station) + glass slides; prepared stomata slides as backup	shared
Digital microscope (one per team)	1 per team (4; two on hand, rotate or borrow two more)
Grid sheets	1 per team (4)
Gloves	optional, shared box at the polish station

YOUR PLAN AND DATA

Clean, complete data is worth as much as a fast result.

Prediction (what will work best, and why): ______________________

Leaf 1 name/type and counts per field (at least 3): ____ ____ ____ **average:** ____

Leaf 2 name/type and counts per field (at least 3): ____ ____ ____ **average:** ____

Leaf 3 name/type and counts per field (at least 3): ____ ____ ____ **average:** ____

Our ranking, water-saving to water-spending (use the averages): ______________________

DEFEND YOUR RANKING

What counts (numbers) support your ranking from water-saving to water-spending? ______________________

Where could counting error sneak in, and how did averaging several fields help? ______________________

HOW YOU SCORE (100 POINTS)

SCORING CRITERION	POINTS
Slide or peel quality	25
Counting accuracy	30
Ranking evidence	25
Team data table	10
Cleanup and care	10
Total	**100**